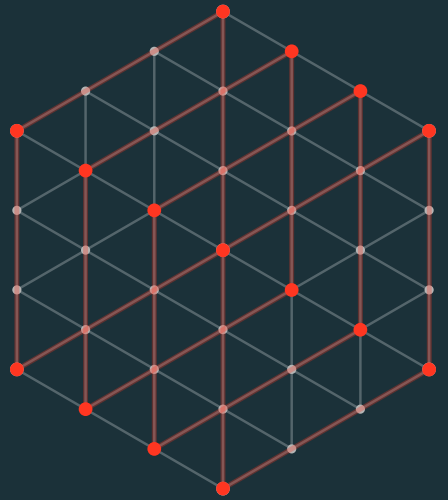
databricks



NORTHSTAR FROZEN FOODS

Stop paying for freight the plan never asked for

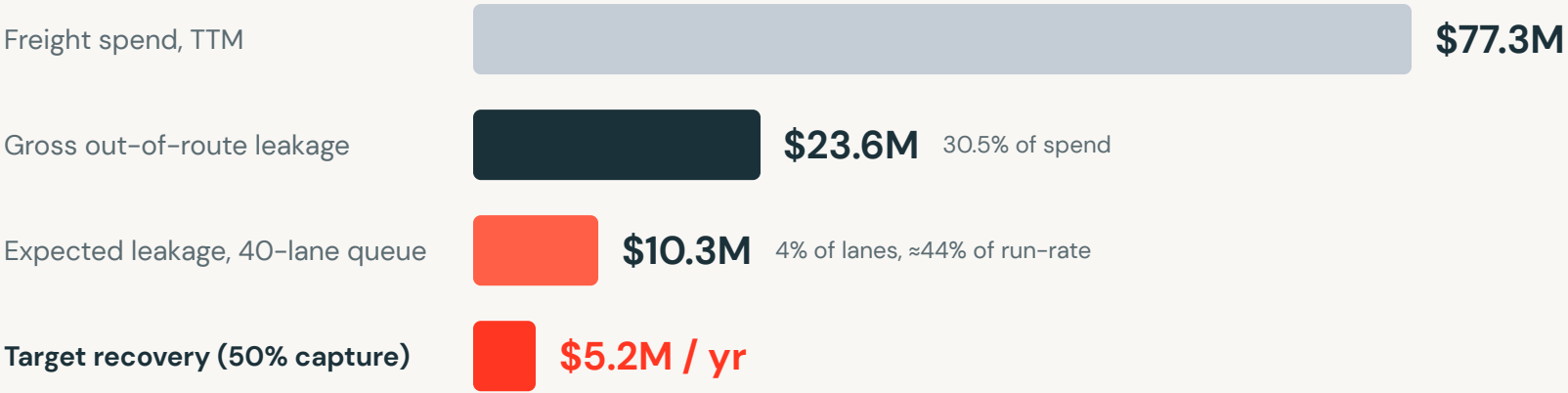
Out-of-route freight leakage · Business case & solution

Prepared for VP Supply Chain & Director, Transportation & Network Planning

Synthetic data

THE OUTCOME

$5.2M a year back from freight, starting with 40 lanes

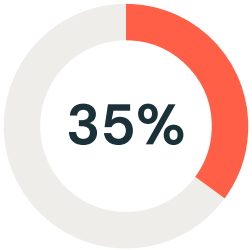


DAILY Flagged the morning after the load moves, not 4–6 weeks later

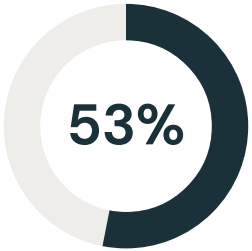
THE ASK 90-day pilot: Drummondville, McAllen & Joliet

THE PROBLEM

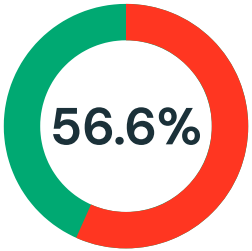
A third of freight runs off-plan, and nobody sees which third costs money



of shipped cases moved on a lane **not in the MRP plan**



of freight spend went through those off-plan lanes



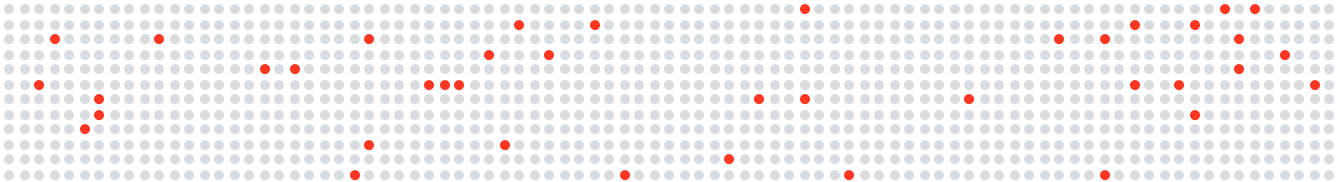
cost more (+$9.03/case). The rest were **cheaper** (-$6.9M)

Signal split across systems:

- TMS: legacy plant codes, broken SKUs
- Invoices in CAD
- Monthly MRP snapshot
- Month-end GL variance

FOR THE DOMAIN OWNER

From 1,068 lanes to a 40-lane queue



01 · Prioritised

ML ranks every lane. The top 20% hold **54%** of leakage.

02 · Explained

GenAI writes the action, owner and $ at stake for each lane.

03 · One click

Approve, reject or snooze. Every decision is logged.

66% of queued $ is a **planning problem**, not a carrier problem: 26 of 40 lanes need an MRP plan fix ($569K / month).

WHAT THE PLANNER SEES

Every lane arrives with an action and a dollar value

Lane #4 REROUTE_TO_PLAN High risk $54,174

Joliet Plant → ~~Atlanta DC~~

Plan: **Columbus DC**

573.9 mi

346.2 mi plan

116 shipments · 142,459 cases · $0.47 / case planned

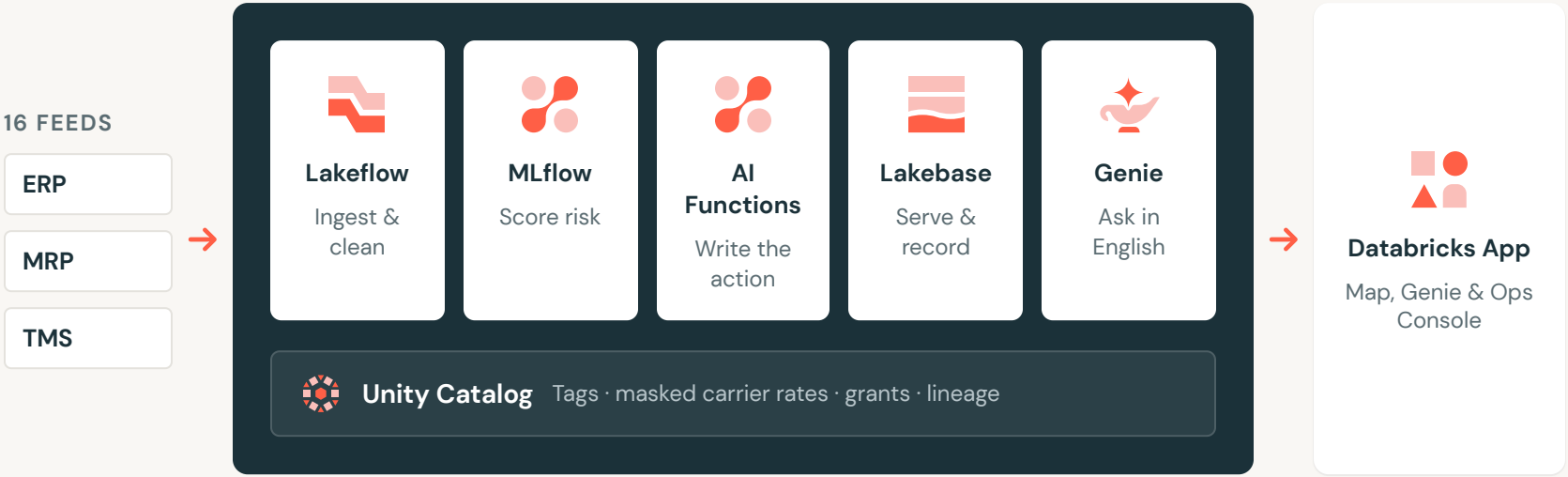
GENAI RECOMMENDATION

"Reroute the 116 out-of-route shipments from Joliet Plant to Columbus DC instead of Atlanta DC, using the planned alternative lane 2507-2551 at $0.47/case."

Approve Reject Snooze

HOW IT WORKS ON DATABRICKS

One governed flow, not six tools



~12 min
One scheduled serverless job

0.917
Model AUC on unseen months

8 / 0
Quality checks per run / rows dropped

2 / 2
Genie answers match hand-written SQL

ROLLOUT PLAN

90 days to a go / no-go on the network

Days 0-30

Connect & baseline

Production extracts, real history, 3 pilot origins

EXIT

Baseline signed · AUC ≥ 0.85

Days 31-60

Go live daily

Planner queue, MRP fix workflow, weekly Finance review

EXIT

≥ 80% of lanes actioned in 7 days

Days 61-90

Prove & extend

Realised vs expected savings, all origins, carrier-rate feed

EXIT

≥ 25% less leakage · go / no-go